PITCH & PRODUCT ARCHITECTURE

Earn it. Prove it. Grow it.

The trusted local work marketplace, verifiable cryptographic work proof (**ProofScript**), and irregular income intelligence (**iBanks**).

Production Application: https://kashwise.netlify.app/login

MARKET REALITY

The Challenge: South Africa's Unrecorded Economy

Millions of skilled workers generate real economic value every day with zero verifiable record.

01

Zero Proof of Work

An artisan completing hundreds of jobs has no portable transcript. Word-of-mouth stays trapped locally; they cannot prove competence to new clients, banks, or landlords.

02

Irregular Income Blindness

Mainstream banking products demand steady monthly payslips. Gig workers experience volatile, clustered cashflows that conventional credit models misinterpret as zero income.

03

Vulnerability & Scams

Unregulated classifieds expose home addresses and phone numbers. Unconfirmed jobs trigger non-payment, predatory middleman fees, and safety risks.

The Opportunity: Turn every completed, client-confirmed job into an immutable cryptographic proof of work, paired with intelligence built for irregular cashflow.

PLATFORM OVERVIEW

The Solution: Three Synchronized Engines

An end-to-end ecosystem connecting local gigs, cryptographic evidence, and financial growth.



1. KashWise Work

LOCAL MARKETPLACE

Match with clients by proximity and proven skill. In-person dual QR handshake confirms on-site physical presence at job start and job completion.

✓ Area-only privacy · No GPS tracking



2. ProofScript

THE EVIDENCE MOAT

Each customer confirmation writes to a living work transcript signed with post-quantum ML-DSA-65. Fully portable JSON records verified in any browser.

✓ FIPS 204 signatures · Public /verify portal



3. iBanks Engine

FINANCIAL INTELLIGENCE

Append-only Postgres ledger built for volatile earnings. Projects 7- & 30-day income forecasts, flags cashflow risks, and powers goal planning.

✓ Earned ≠ Paid out · Honest audit trail

CORE INNOVATION

ProofScript: The Cryptographic Evidence Moat

Turning honest hustle into institutional-grade, portable reputation.

NIST FIPS 204 Standard

Quantum-Resistant Digital Signatures (ML-DSA-65)

Every record is signed server-side by KashWise with modern lattice-based cryptography. Any altered letter, fee, rating or review breaks the 4,412-character signature.

Zero-Knowledge Client Verification

Independent Verification Portal (/verify)

Third-party evaluators (banks, micro-lenders, employers) upload or paste the JSON transcript. Verification executes 100% inside their browser via WebCrypto without phoning home.

Self-Sovereign Work Record

True Worker Portability

Workers are never locked into the platform. One click downloads their entire verifiable history as a standard JSON asset to share anywhere.

kashwise-proofscript-sipho.json SIGNATURE VALID


```
"type": "SideGigsWorkHistory",
"version": 1,
"worker": {
  "name": "Sipho Dlamini",
  "area": "Soweto, Johannesburg",
  "skills": ["painting", "repairs"]
},
"summary": {
  "completed_gigs": 6,
  "average_rating": 4.9
},
"kid": "fdedc4f62e81f105",
"alg": "ML-DSA-65",
"signature": "8f2a...[4,412 chars]"
```


✓ Fully verified live on production at /verify

SECURITY & TRUST

Privacy-First Architecture: Post-Quantum Envelope

Ensuring vulnerable workers and clients never have personal data compromised.

Area-Only Public Profiles

Public profiles and gig listings only show broad neighbourhood centres (e.g., “Soweto” or “Orlando”). Exact street addresses and GPS coordinates are never published.

X-Wing Hybrid Envelope

Phone numbers, real names, and street addresses are encrypted on the user’s device before transmission using ML-KEM-768 + X25519 and AES-256-GCM.

Audited Match Reveal

Decryption only executes server-side when both parties are confirmed matches. Every single contact reveal is logged immutably on the gig’s timeline.

Private Work ID Handshake
When a gig is accepted, KashWise generates a private Work ID (e.g. `KW-7KQ4-M2XP`) known only to the matched client and worker.

POPIA COMPLIANT

FINANCIAL INTELLIGENCE

iBanks: Built for Irregular Income Realities

An event-driven double-entry ledger that transforms financial volatility into clarity.

DOUBLE-ENTRY LEDGER

Earned ≠ Paid Out ≠ Disputed Hold

Postgres double-entry accounting records integer cents with full event provenance. Workers always see their true cash position versus pending settlements.

CONFIDENCE-SCORED FORECASTING

7- & 30-Day Cashflow Ranges

No vague forecasts: iBanks calculates backtest error and confidence bands before projecting upcoming income, helping workers anticipate dry spells.

ACTIONABLE GOAL MATCHING

Earn My Goal (/money/earn-goal)

Set a target (e.g., R900 for Friday rent). The engine shortlists compatible open jobs matching availability, travel radius, and proven skills.



BUSINESS MODEL

Unit Economics & Fair 10% Platform Model

Zero upfront barriers for workers; transparent monetization upon completion.

10%
SUCCESS FEE
Deducted only after client confirms satisfaction.

R0
UPFRONT COST
Free registration, proposal submissions, and service listings.

100%
TRANSPARENT
Both client and worker view exact split before handshake.

PayShap
READY PAYOUTS
Built for rapid settlement into digital wallets and bank accounts.

Sample Transaction Flow: Painting Spaza Shop (Johannesburg)

Client Escrow Deposit (Agreed Total)	R 1,000.00
KashWise 10% Platform Revenue	- R 100.00
Worker Net Payout Take-Home	**R 900.00**
Automated ProofScript Generated	✓ Signed Work Record Added to Worker Passport

TECHNICAL EXECUTION

Production Architecture & Verified Rigor

Cloud-native, serverless, and thoroughly tested across 185+ test suites.

CORE TECHNOLOGY LAYERS

React 19 & Vite 8 SPA
Tailwind CSS 4, Lucide icons, responsive mobile-first shell
`Frontend`

Supabase PostgreSQL with RLS
Row Level Security, Auth, secure file storage, immutable audit logs
`Database`

Netlify Serverless Functions (ESM)
9 TypeScript microservices handling credentials, payments & insights
`API Backend`

@noble/post-quantum Suite
NIST FIPS 204 ML-DSA-65 signatures & ML-KEM-768 envelope encryption
`Crypto`

ENGINEERING QUALITY GATES

- ✓ **185+ Serial Test Suites:** Ledger, crypto, and UI regression passed.
- ✓ **Zero TypeScript Errors:** Strict typechecking with `tsc -b --noEmit`.
- ✓ **Zero ESLint Warnings:** Clean, deterministic production standards.
- ✓ **Release Smoke Suite:** 100% PASS on deployed production domain.

Target Deployed Site: `f8d681ec-ab68-4d61-8a9a-586be62127b7`

STRATEGIC HORIZON

Roadmap: From Working MVP to Scale

A clear path from hackathon validation to nationwide financial inclusion.

PHASE 1 · CURRENT

Production MVP

- Deployed web application on Netlify
- Start/finish QR handshake on site
- NIST FIPS 204 ProofScript generation
- Zero-server browser /verify portal
- Postgres ledger & simulated payouts

✓ Fully Operational & Live

PHASE 2 · Q3 2026

Township Pilot

- Soweto & Tembisa localized artisan cohort
- 50 vetted trade workers (painters, tutors)
- Real PayShap instant payout settlement
- Graphiti local memory optimization
- Field usability and retention validation

⏳ Pilot Architecture Ready

PHASE 3 · 2027

Financial On-Ramps

- Formal bank API integration (FNB, etc.)
- ProofScript accepted as loan collateral
- Micro-insurance opt-in for gig injuries
- Native iOS & Android store release
- Nationwide expansion across 9 provinces

🚀 Scalable Inclusion Network

LIVE EVALUATION READY

Experience KashWise Today

Log in to explore the live marketplace, test the in-person QR handshake, download Sipho's post-quantum ProofScript, and verify it in your browser.

Launch Application Login →

Direct URL: https://kashwise.netlify.app/login